

GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

$f_{GHz}^{(GT)}$: [row 8] GT Freq (GHz)

$|XeCU|$: [row 26] XeCU count

$|XeCore|_{perXeCU}$: [row 27] XeCore per XeCU

$|EU|_{perXeCore}$: [row 28] EU per XeCore

$|L2Banks|_{perXeCU}$: [row 29] L2 banks per XeCU

$L2BankSize_{MB}$: [row 30] bank capacity (MB)

D_{DPAS} : [row 32] DPAS depth

$\eta_{systolic}$: [row 35] compute efficiency %

$L1RdBW_{Bp(Clk \cdot EU)}^{max}$: [row 86] L1 read max (B/EU/clock)

$L1WrBW_{Bp(Clk \cdot EU)}^{max}$: [row 87] L1 write max (B/EU/clock)

$GtiRdBW_{BpClk}^{max}$: [row 97] GTI read max BW (B/clock)

$GtiWrBW_{BpClk}^{max}$: [row 98] GTI write max BW (B/clock)

$MemBW_{GBps}^{max}$: [row 124] max possible HBM BW (GB/s)

$DPAS_{MixFmt}$: [row 1000] native DPAS mixed-precision support

Workload Parameters (Pre-Defined)

$Fmt^{(matA)}$: [row 3] input A data format (e.g. fp8)

$Fmt^{(matB)}$: [row 4] input B data format (e.g. fp4)

$Fmt^{(matD \downarrow)}$: [row 5] output D data format (e.g. fp8)

M_{dim} : [row 9] M dimension

K_{dim} : [row 10] K dimension

N_{dim} : [row 11] N dimension

$Byte_{perElement}^{(matD)}$: [row 24] output bytes per element (fp32)

ρ_{GT} : [row 34] machine occupancy %

$M_{perThread}$: [row 39] M per thread

$K_{perThread}$: [row 40] K per thread

$N_{perThread}$: [row 41] N per thread

$W_{thread}^{(ThreadGroup)}$: [row 45] TG width in units of thread

$H_{thread}^{(ThreadGroup)}$: [row 46] TG height in units of thread

$W_{TG,keep cluster size as 4}^{(XeCoreCluster)}$: [row 50] XeCore cluster width in units of TG (keep cluster size as 4)

$H_{TG,keep cluster size as 4}^{(XeCoreCluster)}$: [row 51] XeCore cluster height in units of TG (keep cluster size as 4)

$W_{TG}^{(XECUTile)}$: [row 54] XeCU tile width in units of TG

$H_{TG}^{(XECUTile)}$: [row 55] XeCU tile height in units of TG

$W_{XeCU}^{(GPUPTile)}$: [row 58] GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\text{Bytes}_{perElement}^{(matA)} = \text{DataFormatToBytes}[\text{Fmt}^{(matA)}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\text{Bytes}_{perElement}^{(matB)} = \text{DataFormatToBytes}[\text{Fmt}^{(matB)}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\text{Bytes}_{perElement}^{(matD\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(matD\downarrow)}]$$

row 33 - EU COUNT

$$|\text{EU}| = |\text{XeCore}|_{perXeCU} \times |\text{EU}|_{perXeCore} \times |\text{XeCU}|$$

row 4002 - MIXED PRECISION DPAS THROUGHPUT DELTA

$$\delta_{\tau_{DPAS}} = \text{DPAS}_{MixFmt} \wedge (\text{Fmt}^{(matA)} \in \{i8\} \wedge \text{Fmt}^{(matB)} \in \{i2, i4\} \vee \text{Fmt}^{(matA)} \in \{i2, i4\} \wedge \text{Fmt}^{(matB)} \in \{i8\})$$

row 4001 - MMA MAC THROUGHPUT PER CHANNEL STAGE

$$\tau_{mMACper(Clk.CHANNEL.STAGE)}^{(peak)} = \frac{4}{\max(\text{Bytes}_{perElement}^{(matA)}, \text{Bytes}_{perElement}^{(matB)})} \times (1 + \delta_{\tau_{DPAS}})$$

row 4000 - MMA MAC THROUGHPUT PER EU

$$\tau_{mMACper(Clk.EU)}^{(peak)} = \tau_{mMACper(Clk.CHANNEL.STAGE)}^{(peak)} \times D_{DPAS} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{mMACper(Clk.XeCore)}^{(peak)} = \tau_{mMACper(Clk.EU)}^{(peak)} \times |\text{EU}|_{perXeCore}$$

row 42 - CLKS PER DPAS

$$\text{CLKS}_{DPAS} = \frac{M_{perThread} \times K_{perThread} \times N_{perThread}}{\frac{\tau_{mMACper(ClK-XeCore)}^{(peak)}}{|EU|_{perXeCore}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$W_{element}^{(Thread)} = N_{perThread}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$H_{element}^{(Thread)} = M_{perThread}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$W_{element}^{(ThreadGroup, realized by multiple MMA iterations)} = W_{thread}^{(ThreadGroup)} \times W_{element}^{(Thread)}$$

row 14 - TG TILES IN N

$$|Tiles|_N^{(TG)} = \frac{N_{dim}}{W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(ThreadGroup)} = H_{thread}^{(ThreadGroup)} \times H_{element}^{(Thread)}$$

row 15 - TG TILES IN M

$$|Tiles|_M^{(TG)} = \frac{M_{dim}}{H_{element}^{(ThreadGroup)}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$W_{element}^{(XeCoreCluster)} = W_{TG, keep cluster size as 4}^{(XeCoreCluster)} \times W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}$$

row 16 - TG CLUSTER TILES IN N

$$|Tiles|_N^{(TG Cluster)} = \frac{N_{dim}}{W_{element}^{(XeCoreCluster)}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCoreCluster)} = H_{TG, keep cluster size as 4}^{(XeCoreCluster)} \times H_{element}^{(ThreadGroup)}$$

row 17 - TG CLUSTER TILES IN M

$$|Tiles|_M^{(TG Cluster)} = \frac{M_{dim}}{H_{element}^{(XeCoreCluster)}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$W_{element}^{(XeCUTile)} = W_{TG}^{(XECUTile)} \times W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}$$

row 18 - XECU TILES IN N

$$|Tiles|_N^{(XeCU)} = \frac{N_{dim}}{W_{element}^{(XeCUTile)}}$$

row 20 - GPU TILES IN N

$$|Tiles|_N^{(GPU)} = \frac{\frac{N_{dim}}{W_{element}^{(XeCUTile)}}}{W_{XeCU}^{(GPUTile)}}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCUTile)} = H_{TG}^{(XECUTile)} \times H_{element}^{(ThreadGroup)}$$

row 19 - XECU TILES IN M

$$|Tiles|_M^{(XeCU)} = \frac{M_{dim}}{H_{element}^{(XeCUTile)}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$H_{XeCU}^{(GPUTile)} = \frac{|XeCU|}{W_{XeCU}^{(GPUTile)}}$$

row 21 - GPU TILES IN M

$$|Tiles|_M^{(GPU)} = \frac{\frac{M_{dim}}{H_{element}^{(XeCUTile)}}}{H_{XeCU}^{(GPUTile)}}$$

row 13 - WAVES

$$N_{waves} = \text{CEILING}(|Tiles|_N^{(GPU)}, 1) \times \text{CEILING}(|Tiles|_M^{(GPU)}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$W_{element}^{(GPUTile)} = W_{XeCU}^{(GPUTile)} \times W_{element}^{(XeCUTile)}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$H_{element}^{(GPUTile)} = H_{XeCU}^{(GPUTile)} \times H_{element}^{(XeCUTile)}$$

row 63 - MAT A INPUT SIZE B

$$\text{Size}_B^{(matA)} = M_{dim} \times K_{dim} \times \text{Bytes}_{perElement}^{(matA)}$$

row 64 - MAT B INPUT SIZE B

$$\text{Size}_B^{(matB)} = K_{dim} \times N_{dim} \times \text{Bytes}_{perElement}^{(matB)}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\text{Size}_B^{(matC, matD)} = M_{dim} \times N_{dim} \times \text{Bytes}_{perElement}^{(matD\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\text{Size}_B^{(matD\downarrow)} = M_{dim} \times N_{dim} \times \text{Byte}_{perElement}^{(matD)}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$\text{L2Size}_B = \text{L2BankSize}_{MB} \times |\text{L2Banks}|_{perXeCU} \times 1024 \times 1024$$

row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF THREAD DIVERGENCE

$$|\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in } L2)} = \min(20000 \times \frac{K_{perThread}}{\text{CLKS}_{DPAS}}, K_{dim})$$

row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE INSTANCE AND SINGLE WAVE

$$\begin{aligned} \text{TotalRequiredL2Size}_{matB\text{ always hit}}^{1\text{ instance, 1 wave}} &= H_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matA)} \times |\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in } L2)} \\ &+ W_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matB)} \times |\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in } L2)} \\ &+ W_{element}^{(XeCUTile)} \times H_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matD\downarrow)} \end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\text{MemBW}_{BpClk}^{max} = \frac{\text{MemBW}_{GBps}^{max}}{f_{GHz}^{(GT)}}$$

row 5000 - WORKLOAD MAC PER XECORE

$$\text{WL}_{MAC}^{(XeCore)} = \frac{M_{dim} \times K_{dim} \times N_{dim}}{|\text{XeCore}|_{perXeCU} \times |\text{XeCU}|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$T_{clk}^{(total)} = \frac{\text{WL}_{MAC}^{(XeCore)}}{\tau_{mMACper(Cl k \cdot XeCore)}^{(peak)} \times \eta_{systolic}}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned} \text{L2Rd}_B^{(total)} = & \text{Size}_B^{(matA)} \times \text{CEILING}\left(\frac{N_{dim}}{W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}}, 1\right) \\ & + \text{Size}_B^{(matB)} \times \text{CEILING}\left(\frac{M_{dim}}{H_{element}^{(ThreadGroup)}}, 1\right) \end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$\text{L2Wr}_B^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 71 - TOTAL L1 READ B

$$\begin{aligned} \text{L1Rd}_B^{(total)} = & \text{Size}_B^{(matA)} \times \text{CEILING}\left(\frac{N_{dim}}{W_{element}^{(Thread)}}}, 1\right) \\ & + \text{Size}_B^{(matB)} \times \text{CEILING}\left(\frac{M_{dim}}{H_{element}^{(Thread)}}}, 1\right) \end{aligned}$$

row 72 - TOTAL L1 WRITE B

$$\text{L1Wr}_B^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 74 - L2 READ B XECORE CLK

$$\text{L2RdBW}_{Bp(Cl k \cdot XeCore)} = \frac{\frac{\text{L2Rd}_B^{(total)}}{|\text{XeCU}| \times |\text{XeCore}|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 75 - L2 WRITE B XECORE CLK

$$\text{L2WrBW}_{Bp(Cl k \cdot XeCore)} = \frac{\frac{\text{L2Wr}_B^{(total)}}{|\text{XeCU}| \times |\text{XeCore}|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\begin{aligned} \text{L2RdWrBW}_{Bp(Cl k \cdot XeCore)} = & \text{L2RdBW}_{Bp(Cl k \cdot XeCore)} \\ & + \text{L2WrBW}_{Bp(Cl k \cdot XeCore)} \end{aligned}$$

row 79 - L1 READ B EU CLK

$$\text{L1RdBW}_{Bp(Cl k \cdot EU)} = \frac{\frac{\text{L1Rd}_B^{(total)}}{|\text{EU}|}}{T_{clk}^{(total)}}$$

row 77 - L1 READ B XECORE CLK

$$\text{L1RdBW}_{Bp(Cl k \cdot XeCore)} = \text{L1RdBW}_{Bp(Cl k \cdot EU)} \times |\text{EU}|_{perXeCore}$$

row 80 - L1 WRITE B EU CLK

$$L1WrBW_{Bp(Clk \cdot EU)} = \frac{\frac{L1WrB^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

row 78 - L1 WRITE B XECORECLK

$$L1WrBW_{Bp(Clk \cdot XeCore)} = L1WrBW_{Bp(Clk \cdot EU)} \times |EU|_{perXeCore}$$

row 83 - L2 READ MAX B XECORE CLK

$$L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)} = \frac{|XeCore|_{perXeCU} \times 64}{|XeCore|_{perXeCU}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{L2RdBWpXeCore} = \frac{L2RdBW_{Bp(Clk \cdot XeCore)}}{L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{L2WrBWpXeCore} = \frac{L2WrBW_{Bp(Clk \cdot XeCore)}}{L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{L2RdWrBWpXeCore} = \frac{L2RdWrBW_{Bp(Clk \cdot XeCore)}}{L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{L1RdBWpEU} = \frac{L1RdBW_{Bp(Clk \cdot EU)}}{L1RdBW_{Bp(Clk \cdot EU)}^{max}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{L1WrBWpEU} = \frac{L1WrBW_{Bp(Clk \cdot EU)}}{L1WrBW_{Bp(Clk \cdot EU)}^{max}}$$

**row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING
DATA SET PCT**

$$\text{L2HitRate}_{\text{random } WS \text{ access}} = \min\left(\frac{\text{L2Size}_B}{\text{TotalRequiredL2Size}_{\text{1 instance, 1 wave}}^{\text{matB always hit}}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$\text{L2MissRate} = 1 - \text{L2HitRate}_{\text{random } WS \text{ access}}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$\text{L2RdB}_{\text{total}} = \text{L2Rd}_B^{(\text{total})}$$

**row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE
PCT**

$$\text{P}_{\text{L2Hit, after 1st wave}}^{(\text{matA})} = \text{IF}\left(\left(\text{Size}_B^{(\text{matA})} + \text{Size}_B^{(\text{matB})} + \text{Size}_B^{(\text{matC, matD})}\right) \leq \text{L2Size}_B, 1.0, \text{IF}\left(\text{K}_{\text{dim}} > 2 \times |\text{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})} - \frac{\text{K}_{\text{dim}} - |\text{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})}}{|\text{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})}}\right)\right)$$

**row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE
PCT**

$$\text{P}_{\text{L2Hit, after 1st wave}}^{(\text{matB})} = \text{IF}\left(\left(\text{Size}_B^{(\text{matA})} + \text{Size}_B^{(\text{matB})} + \text{Size}_B^{(\text{matC, matD})}\right) \leq \text{L2Size}_B, 1.0, 0.0\right)$$

**row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE
PCT**

$$\text{P}_{\text{L2Miss, after 1st wave}}^{(\text{matA})} = 1 - \text{P}_{\text{L2Hit, after 1st wave}}^{(\text{matA})}$$

**row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE
PCT**

$$\text{P}_{\text{L2Miss, after 1st wave}}^{(\text{matB})} = 1 - \text{P}_{\text{L2Hit, after 1st wave}}^{(\text{matB})}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER COLD CACHE

$$\text{MemRdB}_{cold\ start}^{(wave)} = \frac{\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}}}{2}$$

$$\begin{aligned} \left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}} &= \text{Size}_B^{(matA)} \\ &+ \text{Size}_B^{(matA)} \times \left(\text{CEILING}\left(\frac{N_{dim}}{W_{element}^{(XeCUTile)}}, 1\right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matA)} \\ &+ \text{Size}_B^{(matB)} \\ &+ \text{Size}_B^{(matB)} \times \left(\text{CEILING}\left(\frac{M_{dim}}{H_{element}^{(XeCUTile)}}, 1\right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matB)} \\ &+ \left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}} \right)_{\text{aux4}} \times \text{L2MissRate} \end{aligned}$$

$$\begin{aligned} \left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}} \right)_{\text{aux4}} &= \text{L2RdB}_{total} \\ &- \left(\left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}} \right)_{\text{aux4}} \right)_{\text{aux1}} \end{aligned}$$

$$\begin{aligned} \left(\left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{\text{num}} \right)_{\text{aux4}} \right)_{\text{aux1}} &= \text{Size}_B^{(matA)} \\ &+ \text{Size}_B^{(matA)} \times \left(\text{CEILING}\left(\frac{N_{dim}}{W_{element}^{(XeCUTile)}}, 1\right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matA)} \\ &+ \text{Size}_B^{(matB)} \\ &+ \text{Size}_B^{(matB)} \times \left(\text{CEILING}\left(\frac{M_{dim}}{H_{element}^{(XeCUTile)}}, 1\right) - 1 \right) \times P_{L2Miss, after\ 1st\ wave}^{(matB)} \end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\text{MemWrB}^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 118 - TOTAL HBM B

$$\text{MemRdWrB}^{(total)} = \text{MemWrB}^{(total)} + \text{MemRdB}_{cold\ start}^{(wave)}$$

row 121 - HBM READ B CLK

$$\text{MemRdBW}_{BpClk} = \frac{\text{MemRdB}_{cold\ start}^{(wave)}}{T_{clk}^{(total)}}$$

row 122 - HBM WRITE B CLK

$$\text{MemWrBW}_{BpClk} = \frac{\text{MemWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 123 - HBM TOTAL B CLK

$$\text{MemRdWrBW}_{BpClk} = \frac{\text{MemRdWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 126 - HBM BW PCT

$$\eta_{MemBW} = \frac{\text{MemRdWrBW}_{BpClk}}{\text{MemBW}_{BpClk}^{max}}$$

row 127 - GTI READ BW PCT

$$\eta_{GTIRdBW} = \frac{\text{MemRdBW}_{BpClk}}{\text{GtiRdBW}_{BpClk}^{max}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{GTIWrBW} = \frac{\text{MemWrBW}_{BpClk}}{\text{GtiWrBW}_{BpClk}^{max}}$$

DataFormatToBytes

Reference mapping used for workload data-format to bytes-per-element conversion.

```
DataFormatToBytes = {
    'fp16': 2,
    'fp4': 0.5,
    'fp6': 0.75,
    'fp8': 1,
    'i2': 0.25,
    'i4': 0.5,
    'i8': 1,
    'mxfp4': 0.53125,
    'mxfp6': 0.78125,
    'mxfp8': 1.03125,
    'mxi8': 1.03125,
    'nvfp4_4x': 0.5625,
    'nvfp4_6x': 0.5625,
    'tf32': 4,
}
```